4 Architectural

4.1 Summary

The architectural design scope is limited to work in and around the Generator room in Building 30 for this project. All work will be designed to the applicable codes and standards referenced in Section 3 of this narrative.

4.2 Project Description - Building 30 (ACC)

During design for this project, the Fire Protection Engineering analysis identified a deficiency in the existing generator room regarding interior partitions constructed from an acoustic concrete masonry unit (cmu), also known as SoundBlox. This masonry unit, in an 8" thickness, is typically constructed as an unreinforced non-fire rated partition, but can be fabricated with a second cell, allowing the block to be reinforced and grouted, achieving a 1-hr fire rated partition. The only way to determine for sure which type exists in the facility would be destructive testing.

Current NFPA 110-2025 section 7.2.1.1. requires 2-hour fire resistance rated barriers for Emergency Power Supply Rooms. This document applies primarily to new installations of Emergency and Standby Power Systems. The VA determined that interior partitions should be upgraded to comply with the current code requirement.

The extent of wall mounted conduits and equipment made constructing a new partition inside the Generator room problematic. Doing so would also eliminate the sound reduction capability of the CMU. The design team (with VA concurrence) elected to construct the new fire-rated partition on the exterior sides of the two interior partitions making up the room. Constructing the partition against the masonry required the use of a shaft wall assembly allowing all components to be installed from one side. One difficulty is the existing CMU partition at the adjacent Fire Pump room prevents installing continuous shaft wall separation around the room. GDM is proposeding a shaft wall extension on both sides of the CMU to provide fully sealed overlapping protection in lieu of continuity at the corner of Generator room. This proposal was accepted by the VA.

One final hurdle was discovered in an as-built drawing discrepancy for the exterior door pair location in Mech/Fire Protection Room 1A29. Existing door frame is installed too close to the CMU wall at Generator preventing construction of the new fire rated partition through the exterior wall (as required by code). VA determined the best solution to be replacement with a new smaller exterior door and frame allowing proper partition construction / termination through exterior wall stud cavity.

Analysis indicates the existing lightweight concrete floor/metal deck and 5/8" gypsum board ceiling above the Generator room appears to exceed the 2-Hour separation requirement.

4.3 Construction Tasks

- Provide new 2-hour fire-rated wall assembly separating existing interior CMU partitions of the Generator room 1A28 from adjacent rooms Office 1A20, Corridor C1-05, and Mech/Fire Protection Room 1A29.

- Corridor C1-05, Office 1A20
  - Remove, safely store and reinstall existing handrails, crash rails, photographs and furniture.
  - Remove, safely store & reinstall existing ceiling grid, wall angles and ceiling tiles.
  - Remove and reinstall all surface mounted conduit, receptacles, exit sign, and/or junction boxes.
  - Remove existing gypsum board wall finish, acoustic insulation and metal stud framing.
  - Construct new continuous 2-hour fire rated shaft wall assembly floor to ceiling. and finish to match existing.
  - Seal all wall penetrations through the new partition.
  - Finish exposed surfaces to match existing materials and colors.
  - Restore floor and ceiling finishes.

- Mechanical/Fire Protection Room 1A29
    - Contractor to remove all wall mounted conduit, devices and equipment required for new partition installation.
    - Provide safe storage and/or temporary support for conduit, devices and equipment.
    - Remove exterior door pair and frame.
    - Construct new continuous 2-hour fire rated shaft wall assembly and tape for smooth finish.
    - Seal all wall penetrations through the new partition.
    - Paint new partition finishes to match adjacent room wall color.
    - Provide new exterior louvered door pair and frame.
    - Install new rubber wall base sealed to floor for moisture protection during testing.
    - Reinstall conduit, devices and equipment on new partition after painting and wall base are complete.

5 Electrical Engineering Design Narrative

5.1 Building T-1 (E Wing)

- Replace existing 200KW diesel powered generator with new 200KW generator. The generator will be provided with an aluminum enclosure and a stainless-steel tank to provide corrosion resistance necessary for the local climate conditions. Tank will exceed the required capacity of 96-hour operation at approximately 50% load.
    - The peak demand load is 90.84KW. The standard factory size tank is 777 gallons. The generator would operate at approximately 7.54 gal/hour at the measured peak load. This would mean that at 90.84KW, the generator runtime would be approximately 103.05 hours.
    - The 96 hours is due to the main computer room (MCR) being located in the building, Since the building is typically only occupied during normal business hours, and the MCR is approximately 25% of the demand load, the standard 777 fuel tank could provide an estimate of 200 hours of continuous emergency power.
    - The standard fuel tank will allow for the generator to mount in the existing footprint of the replacement slab without increasing the foot print size of the slab.
    - A custom tank would be required for the full load rating of a 200KW generator and would increasing the delivery date and costs considerably.
- A tap box will be provided to inner tie the new generator with the existing ATS to comply with NEC 700 requirements and will include the following:
    - An internal manual transfer switch (MTS) allows connection for temporary generator connection, including battery connection and engine start contact for a portable generator.
    - Connection for annual load bank testing.
    - The generator and engine start controls would be routed to the tap box.
- Provide provisions to monitor the new generator on the existing DDC system (refer to Mechanical documents).
- A temporary generator will be provided prior to removal of existing generator and be provided with 96 hours of fuel onsite. The temporary power requirements will be outlined in the construction drawings.
- The existing generator feeder includes power and control wiring within the same conduit, which voids the UL2200 listing of the generator. The feeder also routes thought the existing switchgear and penetrates into the electrical room Normal and emergency power feeders cannot route through common enclosures per NEC 240. Since a new tap box is being installed on the west wall of Bldg. T-1, all conduits will need to be routed from the generator to the new tap box. Utilizing the existing conduit or conduit pathway is not feasible or conducive to the design.
- All new power and control conduits from the generator will be installed in a common trench to the tap box. There will be a spare conduit and a conduit for the DDC system that will run from the generator and penetrate into the crawl space below the tap box.
- The clearance of the existing generator and switchgear does not comply with NEC Table 110-34. The generator is less than 4 ft from the 12,470V transformer. Per NEC Table 110-34, Condition 1, the clearance requirement is 5ft. In order for the new slab to maintain existing slab dimensions, the existing fence will receive double wide gate to accommodate the access to the generator from the westside.

5.2 Building 30 (ACC)

Existing 750KW generator to remain.

A temporary generator and distribution will be installed prior to construction of upgrading the emergency distribution and fuel delivery system. The temporary distribution will be connected to the emergency side of the ATSs. Temporary generator will be provided with 24 hours of fuel and will be outlined in the drawings.

Fire Pump Room rating will be upgraded to a 2-hour fire rating which will require relocation of all electrical conduits, panels, and equipment on the adjacent wall between to the generator room:

- Fire pump ATS to be temporarily relocated in fire pump room and reconnected to the normal power source and the temporary generator source.
- Irrigation wiring and DDC controls and all associated conduits and wiring to be removed off the wall and reestablished after the upgrade is complete.
- Fire pump to be relocated to permanent location after the fire pump room is completed. Normal power source connection to be established. Temporary power must be reestablished until all work associated with the generator switchboard is complete.

New interior of existing generator switchboard will be retrofitted into the existing switchboard enclosure to accommodate separate main circuit breakers for the emergency feed to the fire pump ATS per NEC 240:

- Replace existing generator controls from existing generator switchboard and provide modern control panel withing the incoming section of the generator switchboard.
- Provide new 1200A generator main circuit breaker to feed a new distribution interior that will be reconnected to the emergency feeds of the existing ATS's with the exception of the fire pump ATS (ATS-FP).
- Provide a connection cabinet adjacent to distribution section of the generator switchboard. Includes 1200A breaker to be interlock with generator main. This is to be used for load bank testing and in the event a temporary generator is needed.
- New 50A/3P breaker will be installed on generator to serve the emergency side of fire pump ATS (ATS-FP).

- Existing day tank will be replaced, requiring recircuiting to new 60-gallon return pump (refer to Mechanical documents).
- Provide 480V power feed to new fuel pump system that will be installed adjacent to 5000-gallon main fuel tank. (refer to Mechanical documents). Provide 2" GRC conduits for pump skid, fuel monitoring and a spare from the generator room to a new vault in the fuel tank area. Use the common duct bank with the tap box feeders.
- Existing fuel tank, new fuel piping and pump skid will be grounded per NEC 250.
- The existing 2" conduits are not able to be reused. There will be 3 2" conduits installed for current and future use. See civil for details.

5.3 Building 32A (Parking Garage)

Replace existing 200KW diesel powered generator with new 200KW generator. New generator will be an open unit with a subbase fuel tank with a minimum of 90 minute run time per NFPA 101 requirements.

- New generator will be an open unit with a subbase fuel tank with a minimum of 90 minute run time.
- Generator will be provided with a second circuit breaker for future EV charging system

- A tap box will be provided to innertie the new generator with the existing ATS to comply with NEC 700 requirements and will include the following:
  - An internal manual transfer switch (MTS) allows connection for temporary generator connection, including battery connection and engine start contact for a portable generator.
  - Connection for annual load bank testing

- Provide DDC connection in Buildin 32A that will connect to TR in Building 30, usiting existing spare conduits between buildings.
- Provide provisions to monitor the new generator on the existing DDC system (refer to Mechanical documents).

5.4 Wiring Methods

- General wiring methods are to be in conduit. Minimum conduit size shall be 3/4".
- All conductors shall be copper with type THHN/THWN-2, or XHHW-2 insulation.
- Wiring for power and lighting will be solid conductors for #10 AWG and #12AWG, and stranded conductors for all cables #8 AWG and larger. Minimum size conductors for lighting and power wiring will be #12 AWG.
- Neutral conductors will be full size throughout the system as a minimum. Provide oversized neutrals for telecom distribution. No shared neutrals will be provided.
- The minimum size conductors for control wiring will be #14 AWG.

5.5 Grounding and Bonding

- Grounding for the building will be in compliance with Article 250 of the National Electrical Code. All electrical power and communication systems shall be bonded to the equipment grounding system for both safety and ground potential differences. All branch circuit conduits will contain an equipment ground conductor from panel ground bus to the devices and equipment served.
- The minimum size of any bonding conductor shall be #6 AWG copper.
- All conductor connections to bonding busbars shall be a 2-hole lug connector.

5.6 Raceways

- A complete raceway system will be provided for all lighting and power conductors.
- Rigid metal conduit, intermediate metal conduit and/or electrical metallic tubing will be used within the building. Rigid metal conduit be used where conduit is installed both exposed and in location that is susceptible to damage. Electric metallic tubing may be used in all other interior locations.
- If required, underground conduit will be PVC schedule 40 with Galvanized Rigid Steel bends, concrete encased.
- If used, exposed exterior conduit will be Galvanized Rigid Steel.

6 Mechanical

6.1 Building T-1 (E-Wing)

- Integrate new generator (refer to electrical documents) into facility DDC system for monitoring and alarms.

6.2 Building 30 (ACC)

- Demolish and replace above-grade fuel piping at the main fuel tank and inside the generator room, including the existing day tank and pumps in the generator room.
- Demolish and replace the existing Veeder-Root level monitoring & leak detection panel in the generator room.
- Provide new fuel supply pumps located at the main fuel tank location.
- Provide new 60-gallon day tank inside the generator room, including fuel return pump (day tank to main tank) and secondary containment. GDM recommends a 60-gallon tank capacity in order to avoid triggering the extensive requirements of NFPA 30 section 24 for ‘storage tank buildings’. A 60-gallon day tank is defined as a ‘container’ by NFPA 30-3.3.12 rather than a ‘storage tank’ by NFPA 30-3.3.54.6, which would change the classification of the room, require added ventilation & controls, increased electrical rating requirements, and others.
- Provide new fire damper where existing exhaust duct penetrations generator room wall.
- Integrate existing generator into facility DDC system for monitoring and alarms.

6.3 Building 32A (Parking Garage Generator Building)

- Integrate new generator (refer to electrical documents) into facility DDC system for monitoring and alarms.
- Provide new generator exhaust system (including piping, silencer(s), etc.) for new generator.
- Provide new remote fuel fill system for generator belly tank.
- Provide network extension (fiber or copper) required to establish DDC connectivity in B32A.

6.4 Fuel Oil Systems

- Duplex fuel supply pumps shall be provided at the exterior fuel tank, including automatic controls for pump alternating. Pumps will be installed in outdoor rated enclosure with controls, piping, & valves.
- A 60-gallon day tank assembly with integrated containment will be provided in the generator room. The day tank assembly will also include a return pump for pumping fuel from the day tank to the exterior main fuel tank, and associated piping, valving, and controls.
- Fuel oil piping & fittings will be butt- or socket-welded, except at valves, unions, and tank connections where flanged or threaded fittings are allowed. Adapter fittings will be provided as necessary to connect to existing double-wall piping.
- New fuel piping sections provided inside buildings will generally be single-wall type.
- Double-containment piping systems will be provided at exterior locations near the B30 exterior fuel storage tank wherever possible. Double containment systems will be provided with leak detection.
- New flexible fuel pipe sections will be provided for connection to generator.
- Fuel piping is intended to be located above grade. No underground piping is intended to be provided as part of this project.

- A fuel tank monitoring and alarm system will be provided with the fuel system. The system will monitor fuel tank levels, high & low level alarms, & leak detection.
- The fuel tank monitoring system will also include alarms and sensor information from the day tank.
- The fuel tank monitoring system will include alarm output and other points to DDC system

6.5 DDC Controls

- The existing Trane Tracer Ensemble direct digital control (DDC) system that serves the facility will be updated to integrate the new & existing generators, day tanks, & tank monitoring systems at the 3 buildings in the project scope.
- The DDC head end computer is located in Building T-1 (E-Wing), mechanical shop.
- The existing head end computer will remain in use. The system graphics will be updated to incorporate new and modified mechanical systems provided as part of this project.
- The existing DDC system is primarily BACnet.
- The DDC system will monitor the fuel tank(s), day tank(s), leak detection, and generators, but control & activation of the generator and fuel systems will be per conventional generator operating methodology.

6.6 Mechanical & Plumbing Unknowns/Questions

- Any leak detection located in existing sump near Building 30 fuel storage tank?
    - *CLOSED - Leak detection for the existing underground double-wall piping is a provided using a combined containment configuration near the exterior tank. A standpipe located in the existing concrete vault allows a single leak detection sensor to detect leaks in either the FOS or FOR piping.*
- Is there a second buried or hidden sump for the double containment piping located near the Building 30 exterior fuel tank, adjacent to the existing visible sump that only houses a standpipe for leak detection sensor?
    - *CLOSED - No second sump is present. Refer to above, confirmed by archive photos provided by the VA.*
- Confirmation of suitable DDC panels in scope areas for connection/expansion
    - *CLOSED - Existing DDC panel locations have been confirmed.*
- Existing DDC system drawings?
    - *CLOSED - Existing control drawings & information has been received from Trane.*
- DDC integrator for site?
    - *CLOSED - The VA does not have a specific DDC integrator.*
- DDC on facility OIT LAN/fiber or dedicated controls LAN/fiber?
    - *CLOSED - The facility DDC system appears to operate on the OIT LAN/fiber network.*
- Is a fuel polishing system desired for the B30 genset fuel system?
    - *CLOSED - The VA is currently engaging a contractor to provide fuel polishing services & equipment under a separate contract. No fuel polishing equipment will be provided under GDM's project.*
- Network path for B32A DDC systems? Can new fiber or copper be run from B32A to B32 room P1-101 to B30 TR room 2-D16 via existing ductbank?
    - *CLOSED – A new data connection is being provided from B32A to a TR in Building 30 (3D-116).*

6.7 Mechanical & Plumbing Studies & Calculations

- Pressure drop/pump sizing calculations for fuel pumps.

7 Structural Engineering Design Narrative

7.1 Systems Summary

The structural design of this project is limited to the following items. All items will be designed to the applicable codes and standards referenced in Section 3 of this narrative.

- New concrete equipment pads for the new E-Wing generator and near the existing B-30 fuel tank for the new supply pumps, disconnect and piping supports.
- Foundations for new fence around the new equipment pad at the E-Wing.
- Due to the minimum clearances required around the new generator, the placement of the new generator will be in a different location than the existing generator. A new foundation is required under the new generator, a new slab is required around the new generator. In order not to move other electrical equipment, the new generator requires removal of the existing fence, slab and foundation, and a new enclosure will be constructed. Fencing and slab will be reinstalled in kind (same location and dimensions).
- Seismic anchorage for all new equipment installed on new or existing concrete equipment pads
- Seismic bracing of new suspended utility racks and piping

Per the VA Seismic Design Manual (PG 18-10) § 4.1, all new or relocated nonstructural components, permanent equipment, and their attachments will be designed to the requirements of ASCE 7, Chapter 13. In addition per PG 18-10 § 4.2, since this facility is Seismic Design Category D, permanent equipment and components shall have Special Seismic Certification in accordance with the requirements of ASCE 7 § 13.2.3.

7.2 Design Criteria

The following is a list of the relevant design parameters to be incorporated in the structural design.

GENERAL DESIGN CRITERIA		
Risk Category for Ancillary Facilities	II	Per PG 18-10
Risk Category for Essential Facilities	III	Per PG 18-10

LIVE LOAD CRITERIA		
Mechanical Rooms	150 psf	Per PG 18-10 Table 1

SNOW LOAD CRITERIA		
Ground Snow Load	N/A	

WIND LOAD CRITERIA FOR ANCILLARY FACILITIES		
Basic Wind Speed for Ancillary Facilities	V = 148 mph	Per ASCE Hazards Report
Basic Wind Speed for Essential Facilities	V = 164 mph	Per ASCE Hazards Report
Wind Exposure Category	B	Per ASCE 7

SEISMIC LOAD CRITERIA		
Importance Factor for Ancillary Facilities	$I_e = 1.0$	Per ASCE 7
Importance Factor for Essential Facilities	$I_e = 1.25$	Per ASCE 7
Importance Factor for Critical Equipment	$I_e = 1.5$	Per VA Seismic Design H-18-8
MCE Spectral Acceleration (Short Period)	$S_s = 0.53$	Per VA Seismic Risk Definitions
MCE Spectral Acceleration (1-Second Period)	$S_1 = 0.15$	Per VA Seismic Risk Definitions
5% Damped MCE Spectral Acceleration (Short Period)	$S_{MS} = 0.73$	Per ASCE 7
5% Damped MCE Spectral Acceleration (1-Second Period)	$S_{M1} = 0.42$	Per ASCE 7

Design Spectral Acceleration (Short Period)	S_{DS} = 0.49	Per ASCE 7
Design Spectral Acceleration (Long Period)	S_{D1} = 0.28	Per ASCE 7
Seismic Design Category	D	Per ASCE 7

8 Civil Engineering Design Narrative

8.1 Design Codes and Standards

- Site Development Design Manual - March 2024
- VA Physical Security and Resiliency Design Manual – June 2025
- VA Master Specifications
- Hawaii Department of Transportation
- International Building Code (IBC) – 2021
- PG-18-13 VA Barrier Free Design Guide, May 2025

8.2 Project Description

- Per the VA provided Scope of Work, the existing exterior generator pad will be demolished, removed and replaced for the E Wing building T-1. The pad will be replaced in the same location and same size with the addition of 3 foot extension to the plan south side of the pad to accommodate the tap box location. The existing side walk will be widened to maintain 3' wide walk where the pad was extended. The construction of the new concrete pad will follow the VA Site Development Manual and the physical security requirements.
  Electrical feeders will be installed underground at the E Wing Building including 6 – 1" pvc conduits and 1-4" pvc conduit and will traverse approximately 31 linear feet and core into an existing utility tunnel and fire alarm conduit including 4 -1 " pc conduits and 1 – 4" pvc conduits will traverse to the plan south to the existing utilidor.
- Building 30 has 2 existing 2" conduits in place that have been deemed unusable per a field investigation on July 30, 2025. The conduits will be abandoned in place and three new 2" concrete encased conduits will be routed from the generator room to the exterior fuel tank approximately 153 linear feet. The conduit will be buried at a minimum bury depth of 24" below existing ground using multiple conduits for redundancy and shall exceed 300 feet or three 90 degree bends between manholes or access points.
- Lastly, approximately 35 linear feet of 3-2" pvc conduit and 2 – 1" pvc conduits will be installed between building 32A and 32 to route the new DDC. All underground concrete encasement shall follow APWA Unifor Color Code and will be dyed red. There will also be trace wire/tape installed per DIV 26 05 41.

8.3 Site Preparation

- Site preparation for the new conduits will consist of demolishing and removing existing landscaping, irrigation, hard surfaces such as asphalt and concrete, and other existing site improvements. These improvements will be restored to their existing or better condition upon completion of the new conduit runs and will follow VA standards for earthwork and trenching.

8.4 Site Grading

- There will be limited site grading in areas disturbed by the new conduits and vaults. Site grading will be restored to match existing conditions and VA barrier free act.

8.5 Pavement Section

- It is anticipated that the conduits will disturb existing concrete and asphalt surfaces. Disturbed concrete and asphalt surfaces will be restored to their existing condition or better in accordance with VA specification 32 12 16. All trenches shall be backfilled and compacted appropriately.

9 Fire Protection Engineering Design Narrative

9.1 Codes and Standards

National Fire Protection Association (NFPA)

- NFPA 10 Standard for Portable Fire Extinguishers, 2022
- NFPA 13 Standard for the Installation of Sprinkler Systems, 2025
- NFPA 30: Flammable and Combustible Liquids Code, 2024
- NFPA 37: Standard for the Installation and Use of Stationary Combustion Engines and Gas Turbines, 2024
- NFPA 70 National Electrical Code, 2023
- NFPA 70e Standard for Electrical Safety in the Workplace, 2024
- NFPA 72 National Fire Alarm and Signaling Code, 2025
- NFPA 101 Life Safety Code, 2024
- NFPA 110 Standard for Emergency and Standby Power Systems, 2025
- NFPA 241: Standard for Safeguarding Construction, Alteration, and Demolition Operations, 2022
- NFPA 291: Recommended Practice for Water Flow Testing and Marking of Hydrants, 2025

Department of Veterans Affairs

- VA Fire Protection Design Manual, Ninth, November 1st, 2023

9.2 Compliance with NFPA 101 Life Safety Code, 2024 Edition

Building 30 Generator Room 1-A28

This project qualifies as a Renovation under NFPA 101-2024 Section 43.2.2.

The generator system improvements for Building 30 include an upgrade of the generator room's fuel-delivery system, including new day tank, above-ground piping, pumps, valves, and controls. The existing components to remain include the generator, the exterior 5,000-gallon fuel storage tank, and the underground fuel supply and return piping from the exterior fuel storage tank to the Emergency Generator Room.

Building 30 is an Ambulatory Care Clinic with a use consistent with that of a Business Occupancy per NFPA 101. Building 30 is of Type II (111) construction type per NFPA 220. Generator Room1-A28 is wholly contained within that construction type and the structural fire-resistance remains unchanged.

All new work in Building 30 Generator Room will meet the requirements of existing Business occupancies as outlined in Section 43.4. Existing non-compliant conditions shall not be made more hazardous per Section 43.4.1.4. Modifications are expected to maintain or improve the current level of safety.

Generator Room 1-A28 is required to have a minimum 1-hour fire resistance rated barrier per NFPA 37-2024 Section 4.1.2.1.1**.** Further, NFPA 101-2024 7.2.3.1.12 (3) requires a minimum 1-hour fire resistance rated barrier for generator rooms. However, NFPA 110-2025 7.2.1.1. requires 2-hour fire resistance rated separation for Emergency Power Supply Rooms.

9.3 NFPA 101 Equivalency and Engineering Judgement Letter

An NFPA 101 Equivalency and Engineering Judgment Letter has been prepared to reconcile existing construction conditions with the fire-resistance and separation requirements applicable to the Modernize Generator System project. The package was developed in accordance with NFPA 101-2024 Section 1.4 and provides the technical justification necessary for acceptance by the VA Fire Marshal and the Authority Having Jurisdiction.

The letter addresses three specific equivalencies, each of which was required due to existing building constraints and conditions uncovered during field investigation:

- **Equivalency #1 – Corner Discontinuity of New 2-Hour Shaftwall:**
  Justification for using an overlapped 2-hour shaftwall configuration in lieu of a continuous fire barrier at an existing CMU intersection where continuity cannot be achieved. The proposed system provides equal or greater fire-resistance performance through redundant 2-hour assemblies, fire safing, and rated joint protection.

- **Equivalency #2 – Existing Floor-Ceiling Assembly:**
  Demonstration that the existing structural concrete and suspended Type-X gypsum board ceiling assembly between Generator Room 1-A28 and the pharmacy above provides at least a 2-hour fire-resistance rating, based on ACI/TMS 216.1-14(19) calculations and recognized code provisions.

- **Equivalency #3 – Generator Exhaust Penetration:**
  Engineering justification for a non-standard firestopping condition where the existing insulated 14-inch generator exhaust chimney penetrates a new 2-hour shaftwall. A project-specific Hilti Engineering Judgment (EJ 701251a) provides a 2-hour-rated firestop solution consistent with UL 1479 methodology.

Collectively, the equivalency package demonstrates that all proposed configurations meet or exceed the life-safety intent of NFPA 101 and NFPA 110, preserve required fire-resistance ratings, and maintain continuity of fire and smoke protection for the generator room enclosure. The formal letter requests VA acceptance of these three equivalencies as providing an equivalent or superior level of fire and life safety for this project.

9.4 Compliance with NFPA 37: Standard for the Installation and Use of Stationary Combustion Engines and Gas Turbines, 2024 Edition

In accordance with NFPA 37-2024 Section 4.1.2.1.1, the Building 30 Generator Room 1-A28 shall be enclosed by a one-hour fire-rated barrier. The room is currently protected by a one-hour fire rated barrier.

The day tank shall comply with NFPA 37 Section 6.1 and meet NFPA 30 construction and secondary containment requirements.

Outdoor engine installations shall comply with NFPA 37-2024 Section 4.1.4 by maintaining a clear separation of at least 1.5 m (5 ft) from any door, window, louver, or other wall opening (4.1.4.1) and by being positioned no closer than 1.5 m (5 ft) to structures having combustible exterior walls unless one of the specific allowances is satisfied.

9.5 Compliance with NFPA 30: Flammable and Combustible Liquids Code, 2024 Edition

The day tanks stored within the Building 30 Generator Room 1-A28 individually are classified as containers by NFPA 30-2024. An aggregate of 60 gallons of diesel, a Class II combustible liquid, is intended to be stored within the room. In accordance with NFPA 30-2024 Section 9.6.1, the maximum allowable quantity of diesel fuel permitted within the occupancy is 120 gallons.

9.6 Fire/Smoke and Smoke Dampers

Fire dampers have been specified for the existing air transfer ducts in Generator Room 1-A28 room enclosure.

9.7 Fire Extinguishers

NFPA 10-2022 Table 6.3.1.1 designates Extra Hazard Class B coverage of 80-B rating at 50 ft travel for diesel splash hazard. Existing fire extinguishers meeting this rating and travel distance are provided for Building 30 and Building 32. A new fire extinguisher and exterior cabinet has been specified for Building T-1 E-Wing.

9.8 Fire Stopping of Wall Penetrations

The contractor is required to pull a permit on all fire and/or smoke-rated walls that require new penetration. They will also seal any penetrations in rated walls that are abandoned due to work completed for this scope in accordance with a listed penetration assembly corresponding to the appropriate rating of the barrier being penetrated.

9.9 Fire Alarm and Detection

<u>Building 30 – Generator Room 1-A28</u>

Generator Room 1-A28 is protected by an existing heat detector, model FST-951 W/B210LP BASE. However, per the VA Fire Protection Design Manual, 7.4.E, heat detectors are not required for generator rooms in sprinklered buildings. Further, NFPA 110-2025 does not require a dedicated fire detection system in generator rooms, so removal of this heat detector is code-compliant and will eliminate an unnecessary alarm source. The existing heat detector is currently required to be demolished as part of this design.

- *VAFPDM 7.4.E. Heat detectors are not required unless used in conjunction with elevator shutdown, where used as a substitute for smoke detectors in environments unsuitable for smoke detectors, or where used to protect emergency generators that are not equipped with automatic sprinklers. Exception: Heat detectors are not required in small remote buildings that house emergency generators. Provide heat detectors in all generator rooms in non-sprinklered buildings. The heat detector must be fixed temperature, extra high temperature (325-375 °F) rating. It is anticipated that most generator rooms will be sprinkler protected and will not require heat detectors.*

- There are no fire alarm notification devices serving the generator room. VA Fire Protection Design Manual 7.3. requires visual notification for normally unoccupied rooms greater than 100 ft^2. A new wall mounted speaker strobe has been proposed for Generator Room 1-A28.
- Generator system alarms (trouble, shutdown, fuel spill, generator running, generator fault, generator switch,) will be monitored as supervisory points by the building fire alarm system in accordance with NFPA 101-2024 9.1.3.2.

<u>Building 32 Generator Outbuilding</u>

The generator outbuilding is not currently protected by any fire alarm system, and none is required for this small, unoccupied structure.

GDM proposes generator system alarms (trouble, shutdown, fuel spill, generator running, generator fault, generator switch,) will be monitored as supervisory points by the building fire alarm system in accordance with NFPA 101-2024 9.1.3.2.

A new heat detector has been provided for the Building 32 Generator Room. In accordance with VAFPDM, the heat detector must be fixed temperature, extra high temperature (325-375 °F) rating.

A new heat detector has been provided for Storage PE-101, high-temperature model FST-951H 190°F has been selected due to unconditioned space and expected high ambient ceiling temperatures

Audible / visual notification has been provided in both Storage PE-101 and the Generator Room.

E-Wing Building T-1 – Proposed Course of Action for Fire Alarm
For this exterior generator location, GDM proposes only generator system alarms (trouble, shutdown, fuel spill, generator running, generator fault, generator switch,) will be monitored as supervisory points by the building fire alarm system in accordance with NFPA 101-2024 9.1.3.2.

9.10 Fire Suppression – Fire Sprinklers

Per the VA Fire Protection Design Manual 6.1.D., fire sprinkler systems must be hydraulically calculated by any design approach allowed by NFPA 13, except that the Special Design Approaches identified by NFPA 13 must be used only when specifically permitted by NFPA 101.

Contractors for suppression systems are required to provide calculations and working drawings for new or modified systems. As such, schematic suppression drawings will be provided for this design.

In accordance with the VA Fire Protection Design Manual, all sprinklers are required to be FM approved.

Building 30 – Generator Room 1-A28

- For Generator Rooms, NFPA 37-2024 11.4.5 requires automatic sprinklers to provide a density of 0.3 gpm/ft^2, over the most remote 2500 ft^2, with a maximum sprinkler coverage of 100 ft^2 per sprinkler. This is consistent with the requirements of Extra Hazard Group 1 per NFPA 13-2025 19.2.2.1.
- The room is currently protected by fire sprinkler coverage consistent with that of a Light Hazard fire sprinkler hazard classification (0.1 gpm/ft^2 with a remote area of 1,500 ft^2 and spacing up to 15' between sprinklers.)
- For Generator Room 1-A28, the design intends to demolish the existing light hazard sprinkler coverage and require the construction contractor to provide new hydraulically calculated Extra Hazard Group 1 coverage. This will require new branch lines and the use of upgraded sprinkler heads that are FM Approved, fusible-link type, standard response, and high-temperature rated (286 °F).

Building 30 Hydraulic Calculation Notes:

- Hydraulic calculations are provided to verify water supply only. The fire sprinkler contractor required to provide calculations and working drawings for the modified fire sprinkler system in the generator room.
- Room Design Method: Since the will have 2-hour fire rated separation, the wall ratings will be in excess of the minimum water supply duration, and THE SPRINKLER DESIGN IS eligible to use the Room Design Method per NFPA 13-2025 19.2.3.3. This PERMITS the remote area to be reduced to the footprint of the room (505 SF) per 19.2.3.3.1.
- Reduction in Design Area of Application: Since high temperature sprinklers with a K factor of 11.2 will be used for the generator room, the area of application has been reduced to 2000 SF per NFPA 13 - 2025 19.2.3.2.7.
- An additional flow of 409 GPM has been applied at the branch line point of common connection, Node 15, to increase the overall demand to the minimum required discharge of 0.30 GPM/FT^2 at the reduced area of application of 2000 FT^2 for an applied sprinkler demand of 600 GPM.
- A hose stream allowance of 500 GPM has been added, resulting in a total system demand of 1100 GPM at the point of supply.

Building 30 Hydraulic Calculation Assumptions

The facility is supplied from the site's distribution pumps, however, the hydraulic calculations are conservatively based on a "no pumps on" condition.

The effective point of supply for the hydraulic model is located at the intersection of the 8-inch site water main serving Building 30 and the 6-inch fire service line serving Fire Hydrant No. 4.

The underground piping configuration was modeled based on the March 20, 1997 design drawings. Original design elevations indicate a finish floor elevation of 380.87 feet and a 6-inch fire line elevation of 363.56 feet (net -17.31 feet). The hydraulic model assumes the 8-inch water service is buried approximately 18 feet below grade.

The modeled underground configuration includes an 8-inch to 6-inch reducing tee located upstream of the building entrance. This tee also supplies Fire Hydrant No. 6.

Building 30 Fire Hydrant Flow Test 10/1/2025:

Purpose

Document available water supply to support the hydraulic calculations for the fire sprinkler modifications planned in Generator Room 1-A28 of Building 30.

Personnel

- Testing performed by: National Fire Protection (sprinkler contractor)
- Witnessed By:
  - Conrad Chandler Fire Protection Engineer, GDM
  - Celyn Dezmain, Project Manager, Civil Engineer, GDM
  - Leo Oshiro, COR, VA
  - Kellie Takemoto, Healthcare Engineer, VA

Test Conditions & Method (summary)

- Two hydrant flow tests were conducted in accordance with the requirements of *NFPA 291: Recommended Practice for Water Flow Testing and Marking of Hydrants, 2025 Edition.*
- Since the facility is supported with site distribution pumps, the first test was performed with site pumps off and the second test was recorded with the site pump on.
- Readings recorded:
  - Fire Hydrant #4 (1-11A, 19F): static pressure, residual pressure,
  - Fire Hydrant #5 (1-13, RT12F): flow (gpm).
- The results recorded represent conditions at the test hydrant on the test date and time.

Measured Results:

Test	System Condition	Static (psi) FH #4	Residual (psi) FH #4	Flow (gpm) FH #5
1	Site pump OFF	66	56	843
2	Site pump ON	74	65	939

Calculated Available Flow at 20 psi Residual (NFPA 291 method):

$$\boldsymbol{Q_{20} = Q_r \left(\frac{P_s - 20}{P_s - P_r}\right)^{0.54}}$$

- **Test 1 (pump OFF):** ≈ 1,921 gpm at 20 psi

  - $\boldsymbol{Q_{20}} = \boldsymbol{843}\left(\frac{\boldsymbol{66-20}}{\boldsymbol{66-56}}\right)^{\boldsymbol{0.54}} =$ 1921 gpm
- **Test 2 (pump ON):** ≈ 2,470 gpm at 20 psi
  - $\boldsymbol{Q_{20}} = \boldsymbol{939}\left(\frac{\boldsymbol{74-20}}{\boldsymbol{74-65}}\right)^{\boldsymbol{0.54}} = 2470$ gpm

Test	System Condition	Static (psi) FH #4	Residual (psi) FH #4	Flow (gpm) FH #5	Calculated Available Flow at 20 psi Residual (gpm)
1	Site pump OFF	66	56	843	1921
2	Site pump ON	74	65	939	2470

<u>Building 30 Fire Pump Test 10/1/2025:</u>

Purpose

Document available water supply to support the hydraulic calculations for the fire sprinkler modifications planned in Generator Room 1-A28 of Building 30.

Personnel

- Testing performed by: National Fire Protection (sprinkler contractor)
- Witnessed By:
  - Conrad Chandler Fire Protection Engineer, GDM
  - Celyn Dezmain, Project Manager, Civil Engineer, GDM
  - Leo Oshiro, COR, VA
  - Kellie Takemoto, Healthcare Engineer, VA

Test Conditions & Method (summary)

- Existing Fire Pump Information:

  Pentair Model 3” 1624F
  Type: Electric, Centrifugal Split Case
  Serial 17-2520690
  Impeller Diameter: 6.313”
  Rated Capacity 500 gpm
  Rated Pressure 50 psi
  Max Power 25.2 BHP
  Max Pressure: 25.2 psi
  Max Suction Pressure: 103.5 PSI
  Stages: One
  Pressure @ 150%: 33.9 PSI

- Pump operated at no flow (churn), 100% flow (500 gpm), and 150% rated flow (750 gpm) through the installed test loop and flow-measuring device.
- Testing was performed through the test loop only. A full test-header (open discharge) test was not available due to parked vehicles obstructing the test header.
- Suction and discharge pressures were read at the pump gauges; Net pressure = Discharge − Suction.
- Electrical readings (voltage, amperage) and speed (rpm) recorded at each point.
- One of the 6-inch butterfly valve on the test loop appears impaired. The position indicator is not showing and the valve is potentially not fully opening and fully closing or seat. This condition can affect recirculation and observed pressures and flows, particularly at high-flow points. GDM understands that VA is currently correcting this deficiency outside of this design contract.
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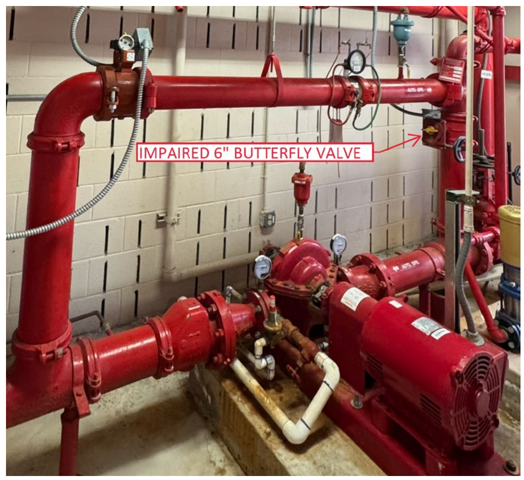

Figure 1 Building 30's Fire Pump and Impaired 6" Butterly Valve

Measured Results 10/1/2025

Point	Flow (gpm)	Suction (psi)	Discharge (psi)	Net Pressure (psi)	Voltage (V)	Amperage (A)	RPM
Churn/No Flow	0	65	120	55	474	15.9	3,578
100% Rated Flow	500	65	117	52	474	25.6	3,595
150% Rated	750	65	105	40	474	28.1	3,542

<u>Comparison with Previously Tested Results</u>

The following table captures and compares the test data from the previous 8/22/2024 and 9/11/2025 fire pump tests and the recently performed 10/1/2025 fire pump test.

Test Point	Test Date	Flow (gpm)	Suction (psi)	Discharge (psi)	Net Pressure (psi)
Churn	**8/22/2024**	0	60	120	60
	9/11/2025	0	65	120	55
	10/1/2025	0	65	120	55
	% Change (from 9/11/2025 Test)	-	0%	0%	0%
100% Rated Flow	**8/22/2024**	533	50	110	60
	9/11/2025	500	68	118	50
	10/1/2025	500	65	117	52
	% Change (from 9/11/2025 Test)	-	-4%	-1%	4%
150% Rated Flow	**8/22/2024**	754	60	106	46
	9/11/2025	750	64	118	54
	10/1/2025	750	65	105	40
	% Change (from 9/11/2025 Test)	-	2%	-11%	-26%

- Suction pressure at the 10/1/2025 test remained relatively consistent with the previous 9/11/2025 test.
- Performance of the net available pressure at the recent 10/1/2025 when compared to the previous 9/11/2025 test:
    - Unchanged at churn.
    - Slightly better at 100% Flow (+4%)
    - Lower at 150% Flow (−26%) driven mainly by a 13 psi drop in discharge.
- It is also worth noting that the pump was recently repacked on August 24, 2025, and there could be a short break-in period during which the packing settles that might slightly affect the pumps performance in speed, amperage, temperature, etc. However, these effects are mechanical and typically small compared with hydraulic behavior. The pump was tested after repacking on September 11, 2025.

Acceptance Comparison – Test Results vs Nameplate and Typical NFPA 20 Criteria.

- Rated head (Nameplate): 50 psi (expected at 500 gpm). Tested 55 psi. Acceptable
- 150% head (Nameplate): 33.9 psi (expected at 750 gpm). Tested 40 psi. Acceptable.
- NFPA-type benchmarks:
    - Churn head:
        - NFPA 20 Benchmark: 101–140% of rated head.
            - *NFPA 20 A.6.2: Shutoff head will range from a minimum of 101 percent to a maximum of 140 percent of rated head.*
        - Tested 55 psi vs rated head of 50psi for 110% of rated head. Acceptable.

- 100% rated flow 500 gpm:
    - Achieved rated flow 500GPM at 52psi, exceeds rated head of 50psi. Acceptable.
- 150% of rated flow 750 gpm:
    - Benchmark: Head ≥65% of rated head (≥32.5 psi for this pump).
        - *NFPA 20 6.2.1: Pumps shall furnish not less than 150 percent of rated capacity at not less than 65 percent of total rated head.*
    - Achieved: 40 psi net which is above NFPA criteria (≥32.5 psi) and the manufacturer's 33.9 psi expectation. This is technically acceptable, however, there has been a decrease in performance in the recent test from the previous 8/22/2024 and 9/11/2025 test. This could be influenced by the suspected impairment of the 6 in. test loop butterfly valve.

Fire Pump Test Conclusion and Recommendations:

- Voltage stable at 474 V.
- Amperage increases with flow as expected: 15.9A to 28.1 A.
- Measured speed near rating of 3560 RPM.
- Pump performance at churn and 100% exceeds the nameplate rating. This is technically acceptable, however, there has been a decrease in performance in the recent test from the previous 8/22/2024 and 9/11/2025 test. This could be influenced by the suspected impairment of the 6 in. test loop butterfly valve.
- The 150% Flow point exceeds the nameplate and NFPA 20 criteria.
- Recommendation: Replace the impaired 6-in. test-loop butterfly valve and position indicator. Tag the valve as impaired until corrected. Perform full fire pump test through test header.

Building 32 Generator Outbuilding – Proposed Course of Action for Fire Sprinklers

- Since the building is a dedicated, detached structure, of non-combustible construction, with a separation of greater than 5 feet from the adjacent building, fire sprinklers are not required per NPFA 37-2024 4.1.2.2. Further, as a Special-Purpose Industrial Occupancy, fire sprinklers are not required by Chapter 40 of NFPA 101-2024. GDM proposes the Building 32 Generator Outbuilding to remain non-sprinklered.

10 Interim Life Safety Measures (ILSM)

Where fire alarm or sprinkler systems are impaired more than four hours in any 24-hour period, the contractor shall provide a documented fire watch. The contractor shall maintain egress routes, protect openings and systems, control hot work, and follow daily combustible waste removal procedures. All ILSMs shall be coordinated with the VA COR and implemented before work commences.

11 Regulated Material Abatement

The VA has shared record information that would indicate a potential for environmental hazards impacting the facility. The wall penetration into building T-1 E-Wing utilidor is expected to contain friable asbestos material. There has been no testing preformed during the design phase due to the extent of disruption to perform the sampling of the utilidor wall for such a small scope of the project. The contractor will be responsible for sampling, testing, and abating / mitigating any hazardous material found at the penetration point. The following are assumed to be an environmental hazard for this project at this time:

- Asbestos
- Confined space involving degassing.